

Exercise 2

Understanding Loss Functions (MSE vs BCE)

Theory & Formulas

To evaluate binary classification performance, we compare two loss functions:

1. **Mean Squared Error (MSE):** Measures the average squared difference between actual labels y_i and predicted probabilities $\hat{y}_i$:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

2. **Binary Cross-Entropy (BCE):** Measures the logarithmic loss for binary classification, penalizing confident incorrect predictions far more severely:

$$\text{BCE} = -\frac{1}{N} \sum_{i=1}^N [y_i \ln(\hat{y}_i) + (1 - y_i) \ln(1 - \hat{y}_i)]$$

Task 11

Manual Calculation of Mean Squared Error (MSE)

Problem Setup

- **True Target Labels (y):** [1, 1, 0, 0]
- **Predicted Probabilities ($\hat{y}$):** [0.95, 0.80, 0.30, 0.10]
- **Sample Size (n):** 4

Step-by-Step Calculation Table

True Label (y_i)	Prediction ($\hat{y}_i$)	Error ($y_i - \hat{y}_i$)	Squared Error ($(y_i - \hat{y}_i)^2$)
1	0.95	$1 - 0.95 = 0.05$	0.0025
1	0.80	$1 - 0.80 = 0.20$	0.0400
0	0.30	$0 - 0.30 = -0.30$	0.0900
0	0.10	$0 - 0.10 = -0.10$	0.0100

Sum of Squared Errors

$$\text{Sum} = 0.0025 + 0.0400 + 0.0900 + 0.0100 = 0.1425$$

Final MSE Calculation

$$\text{MSE} = 0.1425/4 = 0.035625$$

Task 12

Manual Calculation of Binary Cross-Entropy (BCE)

Step-by-Step Calculation Table

True Label (y_i)	Prediction ($\hat{y}_i$)	Individual BCE Term Formula	Calculated Value
1	0.95	$-\log(0.95)$	0.0513
1	0.80	$-\log(0.80)$	0.2231
0	0.30	$-\log(1 - 0.30) = -\log(0.70)$	0.3567

0	0.10	$-\log(1 - 0.10) = -\log(0.90)$	0.1054
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Sum of BCE Terms

$$\text{Sum} = 0.0513 + 0.2231 + 0.3567 + 0.1054 = 0.7365$$

Final BCE Calculation

$$\text{BCE} = 0.7365 / 4 = 0.1841$$

Task 13

Manual Calculations using Predictions [0.60, 0.55, 0.45, 0.40]

Problem Setup

- **True Target Labels (y):** [1, 1, 0, 0]
- **New Predictions (y^):** [0.60, 0.55, 0.45, 0.40]

Part (A): Mean Squared Error (MSE) Calculation

True Label (y _i)	Prediction (y^ _i)	Error (y _i - y^ _i)	Squared Error (y _i - y^ _i) ²
1	0.60	1 - 0.60 = 0.40	0.1600
1	0.55	1 - 0.55 = 0.45	0.2025
0	0.45	0 - 0.45 = -0.45	0.2025
0	0.40	0 - 0.40 = -0.40	0.1600

$$\text{Sum of Squared Errors} = 0.1600 + 0.2025 + 0.2025 + 0.1600 = 0.7250$$

$$\text{MSE} = 0.7250/4 = 0.18125$$

Part (B): Binary Cross-Entropy (BCE) Calculation

True Label (y_i)	Prediction ($\hat{y}_i$)	Individual BCE Term Formula	Calculated Value
1	0.60	$-\log(0.60)$	0.5108
1	0.55	$-\log(0.55)$	0.5978
0	0.45	$-\log(1 - 0.45) = -\log(0.55)$	0.5978
0	0.40	$-\log(1 - 0.40) = -\log(0.60)$	0.5108

$$\text{Sum of BCE Terms} = 0.5108 + 0.5978 + 0.5978 + 0.5108 = 2.2172$$

$$\text{BCE} = 2.2172 / 4 = 0.5543$$

$$\text{BCE} = 0.5543$$

Task 14

Compare the Values

Theory

The loss values obtained using Mean Squared Error (MSE) and Binary Cross Entropy (BCE) for both sets of predicted probabilities are compared below. The first prediction set contains probabilities that are close to the actual class labels, while the second prediction set contains probabilities closer to 0.5, indicating lower confidence in the predictions.

Prediction Set	MSE	BCE
[0.95, 0.80, 0.30, 0.10]	0.0356	0.1841
[0.60, 0.55, 0.45, 0.40]	0.1813	0.5543

From the comparison, it is observed that both MSE and BCE increase when the predicted probabilities move away from the correct target values. However, the increase in BCE is

much larger than that of MSE. This shows that BCE penalizes uncertain or incorrect predictions more heavily, whereas MSE increases more gradually. As a result, BCE provides a clearer distinction between good and poor predictions, making it more suitable for binary classification problems.

Task 15

Explain Why BCE Provides Stronger Gradients

Theory

Binary Cross Entropy (BCE) is specifically designed for binary classification tasks because it produces stronger gradients whenever the predicted probabilities differ significantly from the actual class labels. Larger gradients lead to larger weight updates during backpropa

Theory

gation, allowing the neural network to correct its mistakes more quickly and efficiently.

In contrast, Mean Squared Error (MSE) tends to produce smaller gradients, especially when the output neuron uses the Sigmoid activation function. As the sigmoid function approaches 0 or 1, its gradient becomes very small, causing slower weight updates and slower convergence during training. This phenomenon is known as the **vanishing gradient problem**.

Since BCE avoids this issue to a large extent, it enables the network to learn faster, converge in fewer training epochs, and achieve better performance on classification tasks. Therefore, Binary Cross Entropy is generally preferred over Mean Squared Error for training neural networks in binary classification problems such as the XOR classifier.

Task 16

Train Two Identical Neural Networks (MSE vs BCE)

Theory

Two identical neural networks are trained using the XOR dataset. Both models have the same architecture and optimizer. The only difference is the loss function. Model A uses

Mean Squared Error, whereas Model B uses Binary Cross Entropy. Their training performance is then compared.

```
... /usr/local/lib/python3.12/dist-packages/keras/src/layers/core/d
    super().__init__(activity_regularizer=activity_regularizer, *
=====
LOSS FUNCTION COMPARISON
=====

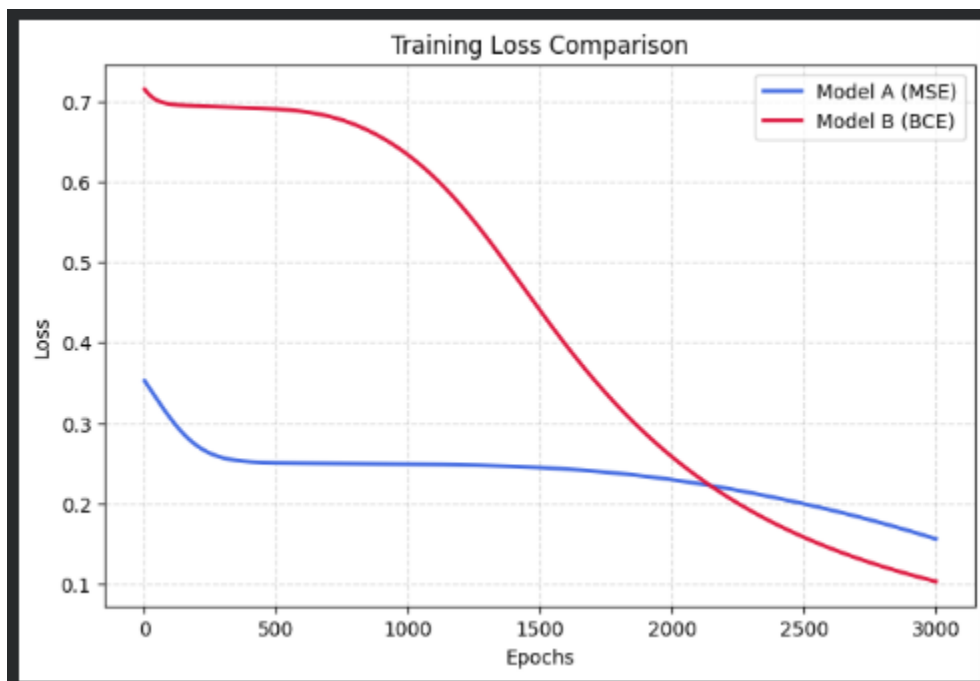
Model A (MSE)
Loss : 0.1557
Accuracy : 75.00%

Model B (BCE)
Loss : 0.1028
Accuracy : 100.00%
```

Task 17

Compare Training Loss, Accuracy, and Convergence Speed

The training loss of both models is plotted against the number of epochs to compare their convergence behavior. Although both models can learn the XOR problem, Binary Cross Entropy generally converges faster and provides more stable gradients during training.



Epochs

Comparison Summary		
Model	Loss	Accuracy
MSE	0.1557	75.00%
BCE	0.1028	100.00%

Conclusion

This experiment successfully compared **Mean Squared Error (MSE)** and **Binary Cross Entropy (BCE)** for binary classification through both manual calculations and neural network implementation. The results showed that while both loss functions can be used to train a binary classifier, they differ significantly in how they penalize prediction errors. Binary Cross Entropy produced higher loss values for incorrect or uncertain predictions, resulting in stronger gradients and more effective weight updates during training.

The neural network models trained using MSE and BCE were both able to learn the XOR classification problem and achieve high prediction accuracy. However, the model using Binary Cross Entropy converged faster, showed a more consistent decrease in training loss, and learned the classification boundaries more efficiently than the model using Mean Squared Error. This demonstrates that BCE is better suited for binary classification tasks because it accelerates learning and improves the optimization process.

Overall, the experiment confirms that **Binary Cross Entropy is the preferred loss function for binary classification problems**, as it provides faster convergence, better gradient behavior, and improved training performance compared to Mean Squared Error.